



EduSuite Pro

System Workflows & Lifecycle Architecture

A comprehensive blueprint documenting the core operations, student lifecycles, examination procedures, and administrative workflows.

DOCUMENT SCOPE:

- Academic Management Lifecycle
- Faculty Operations & HR Workflow
- Examination & Results Processing
- Student Management & Lifecycle Workflow
- Access Control & Admin/Super-Admin Architecture
- Training & Placement Cell Processes
- Library Operations & Member Management

METADATA:

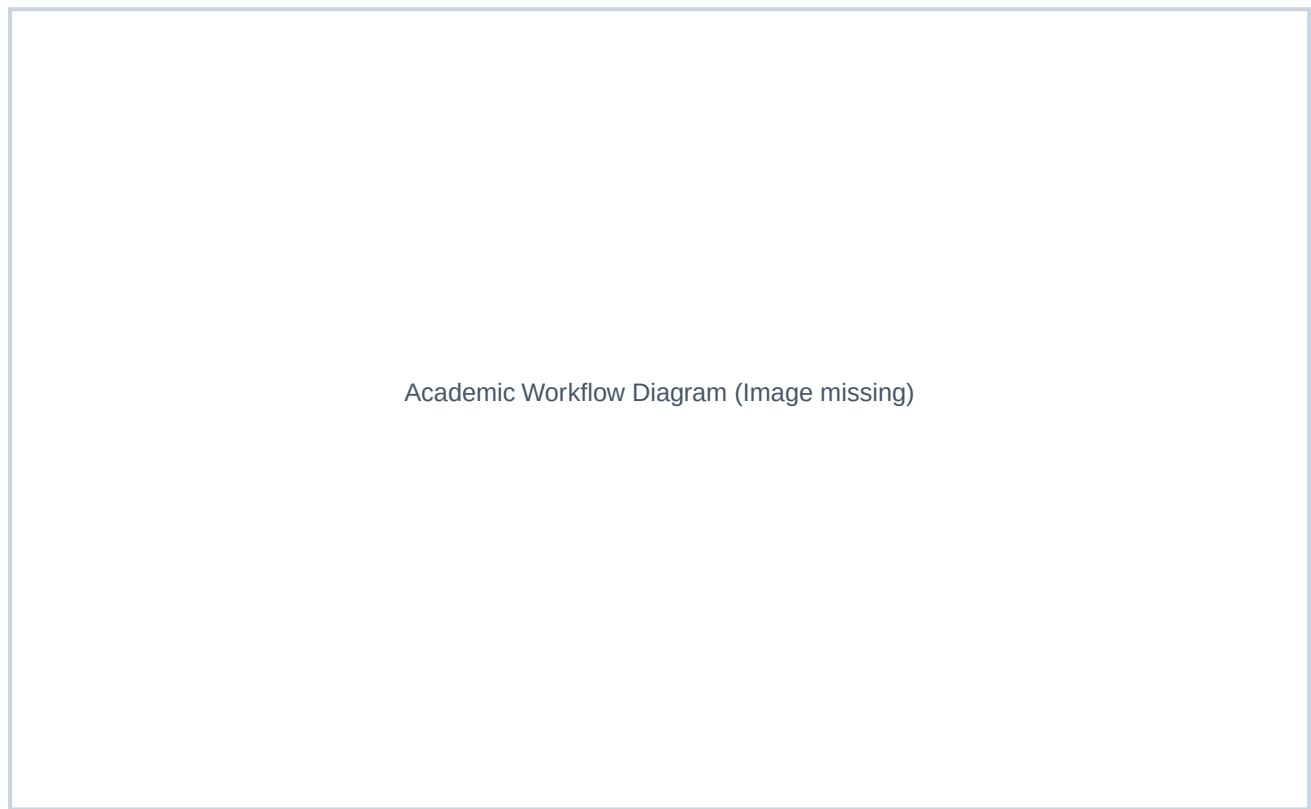
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1. Academic Management Workflow

The Academic Management module drives the curriculum planning, program coordination, and academic scheduling of the institution. It bridges the core rules with operational entities like section mapping, timetable generation, and attendance validation before exams.



Academic Workflow Diagram (Image missing)

2. Faculty Operations & Management

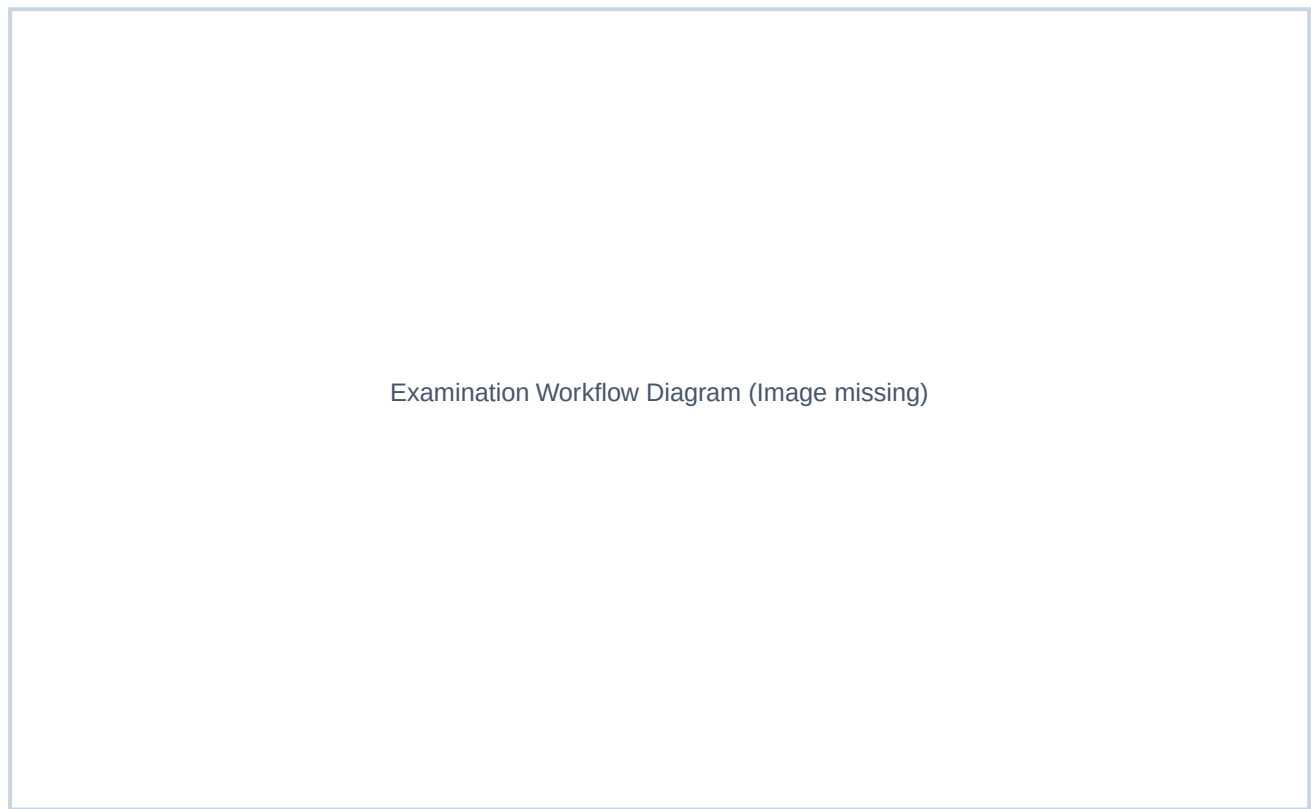
The Faculty Management Workflow defines the onboarding, performance monitoring, class assignment, research tracker, and payroll updates for teaching staff. This module enables automated attendance capture, leave workflow approvals, and promotion reviews.



Faculty Workflow Diagram (Image missing)

3. Examination & Results Processing

The Examination & Results Processing workflow guides the creation of exam schedules, hall ticket approvals, secure internal/external mark entry by faculty members, combined marks aggregation, and final results publication on the student and parent portals.



Examination Workflow Diagram (Image missing)

4. Student Management & Lifecycle

The student lifecycle forms the core database structure of the ERP, mapping a student from pre-admission prospects to alumni archives.

Complete Student Lifecycle Flow

Stage 1: Admission Desk (Pre-Admission / Seat Allocation / Admission)

Stage 2: Student Created & Official ID/Credentials Generated

Stage 3: Master Student Record Linked to Core Modules (Profile, Attendance, Fees, Exams, Services)

Stage 4: Semester Promotion / Branch & Section Transfers

Stage 5: Final Year Completion & Graduating Clearance Checks

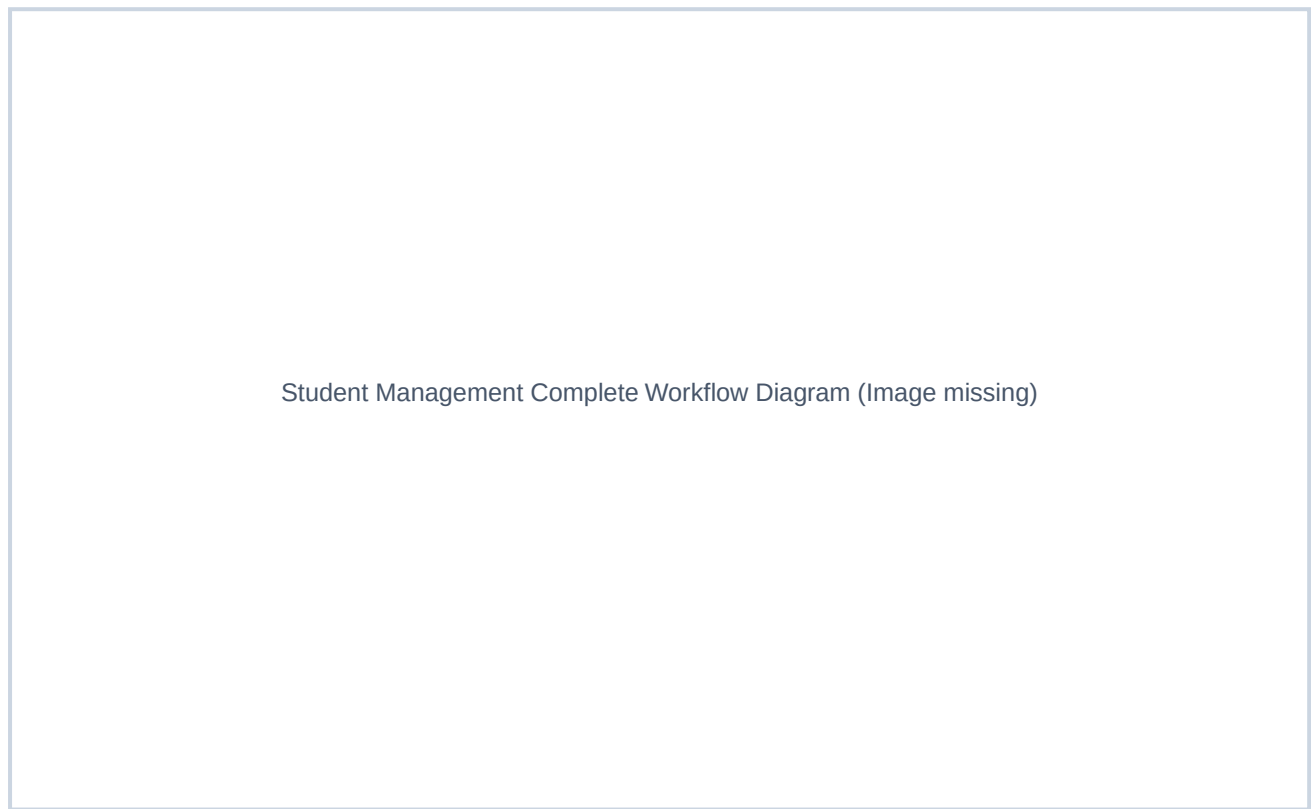
Stage 6: Alumni Registry Auto-Onboarding & Directory Archival

Student Management Module Structure

- **Admission Desk:** Pre-admission screening, Seat Allocation, Document Verification, and ID generation.
 - **Student Profile:** Personal, parent, and medical details, doc repository, and history records.
 - **Attendance Tracker:** Daily check-ins, monthly compliance sheets, and leave approvals.
 - **Examination Portal:** Internal schedules, backlog sheets, hall tickets, and grade cards.
 - **Fees & Finance:** Scholarship matching (JVD support), fee templates, online dues, and receipts.
 - **Campus Services:** Library memberships, Hostel allotments, and Transport/bus route mappings.
 - **Certificates Desk:** Automated generation of Study, Bonafide, Conduct, and Transfer Certificates.
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4.1 Student Management Complete Workflow

This diagram details the comprehensive, end-to-end student lifecycle, illustrating the stages from initial Admission Desk registrations to final Alumni and Archive mappings.



Student Management Complete Workflow Diagram (Image missing)

5. Access Control & Admin/Super-Admin Architecture

EduSuite Pro enforces enterprise-grade security and module control, utilizing Role-Based Access Control (RBAC), responsibility override flags, dynamic feature-flag licensing, and configurable multi-step approval workflows.

Permission Evaluation Formula

Granular permissions are evaluated dynamically in real-time using the following central evaluation equation:

$$\text{Permission} = \text{Base Role} + \text{Responsibility Flags} + \text{Module License} + \text{Action} + \text{Scope}$$

Reorganized Responsibility Flags (Staff Overrides)

Staff privileges are expanded dynamically using categorized Responsibility Flags rather than hardcoded sub-roles:

System Privileges: isSystemAdmin (Global Bypass), isSecurityAdmin, isAuditAdmin, isUserManager.

Academic Privileges: isPrincipal, isVicePrincipal, isDean, isHod (Departmental Head), isExamController, isClassAdvisor, isMentor.

Student Services: isAdmissionsOfficer, isHostelWarden, isTransportOfficer, isPlacementOfficer.

Operations: isFinanceOfficer, isHRManager, isInventoryManager, isLibraryAdmin, isIQACCoordinator.

Configurable Approval Engine

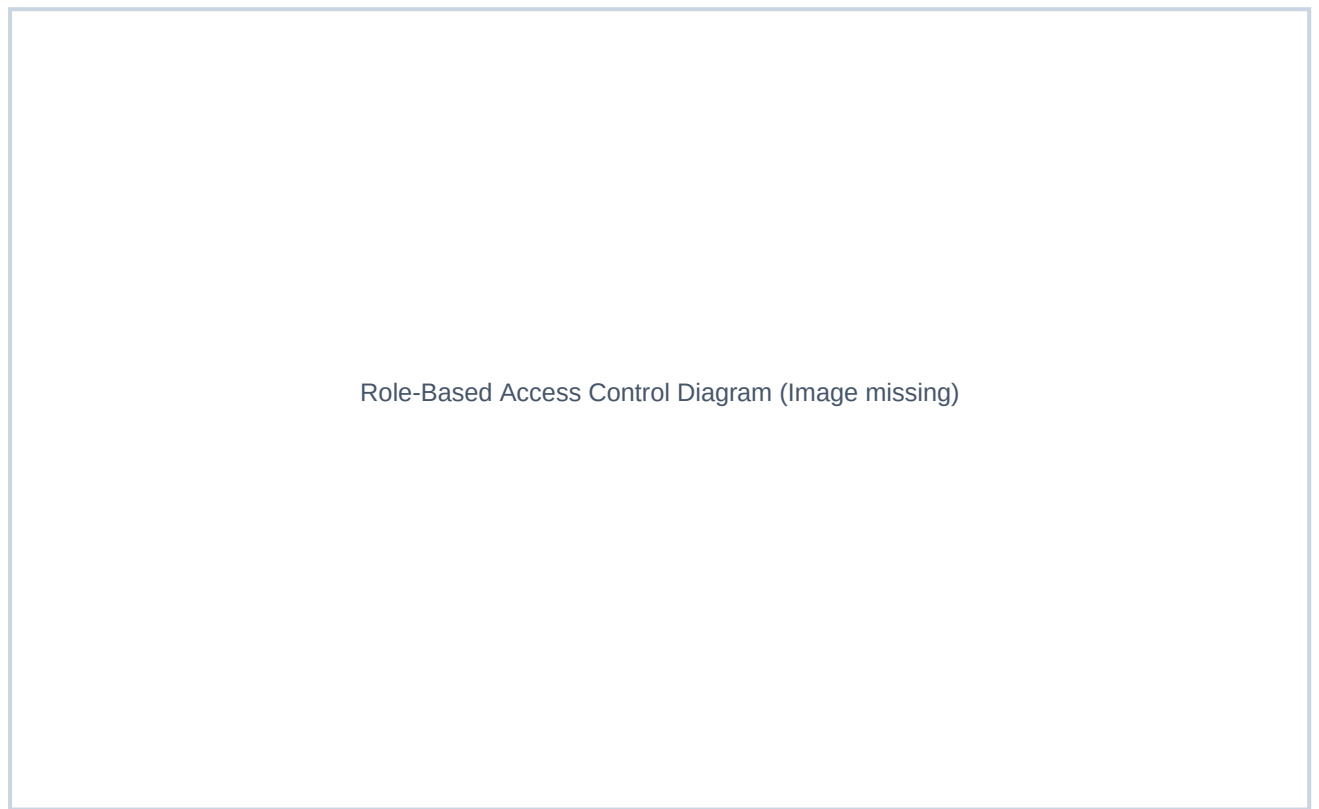
Administrative tasks requiring multi-step authorizations (e.g. Leave approvals, purchases, marks publication) use a sequential approval chain configured in settings. Administrators build step timelines by linking approval checkpoints (e.g., Faculty Advisor -> HOD -> Dean -> Principal).

SaaS Feature-Flagging Engine

SaaS module licensing is tied directly to feature flags. If a license is toggled off: (1) sidebar navigation filters the module, (2) hasPermission blocks backend/frontend routing, and (3) dependent integrations (like the AI Copilot assistant) are disabled.

5.1 Role-Based Access Control Architecture

This diagram details the core 5 login roles, staff privilege override flags, department scopes, module access matrices, and dynamic approval chains.



Role-Based Access Control Diagram (Image missing)

6. Training & Placement Cell Workflow

The Training and Placement (T&P) Cell module handles drive setups, auto-filtering students based on eligibility, assessment creation with dynamic IDEs, and offer letter policies.

Phase 1: Pre-Placement Preparation

- Career Declaration: Students select paths (Placement / Higher Studies / Startup).
- Profile Verification: T&P cell inspects resumes and eligibility credentials.
- Academic Validation: Automatic checks on CGPA, active backlogs, and attendance.

Phase 2: Company & Recruiter Onboarding

- Registration & Verification: Recruiter sign-up and security verification.
- Classification: Segment companies into Mass Recruiter, Core, Dream, and Super Dream tiers.

Phase 3: Drive Management

- Create Drive: Publish role details, salary packages, locations, and schedules.
- Auto-Filtering: System identifies eligible cohorts and triggers sign-up notifications.

Phase 4: Assessment Creation & Approvals

- Assessment Builder: Recruiter constructs tests containing MCQs, Coding IDE (C/C++, Java, Python), or SQL IDE environments.
- Review Queue: Placements Officer inspects, requests revisions, or approves the test for publication.

Phase 5: Assessment Operations & Evaluation

- Setup Test: Schedule date and send access links.
- Test Player: Dynamic full-screen player with code compiler execution (runs code against test cases) and SQL query validator.

Phase 6: Result Moderation & Interview scheduling

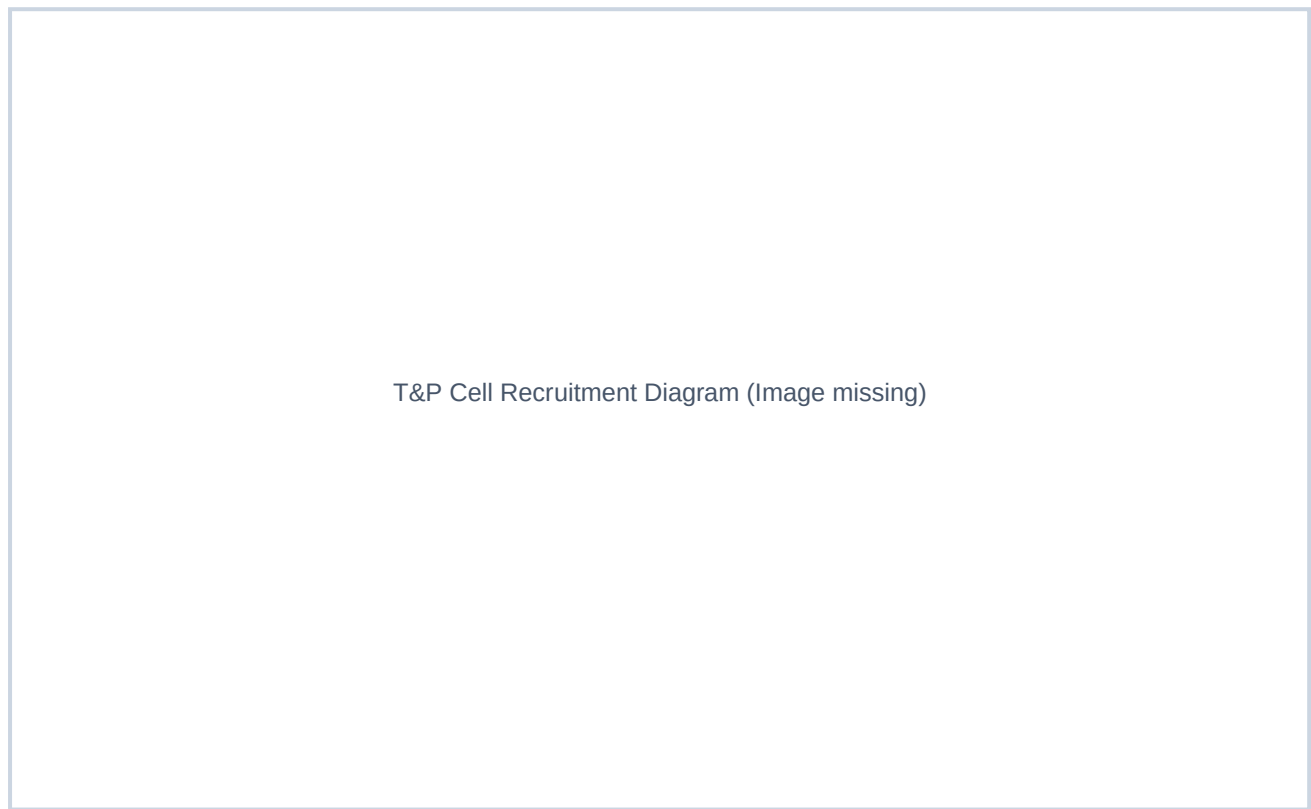
- Audits: Placements Officer audits results and flags. Coordination: Schedule Technical and HR interview rounds.

Phase 7: Offer & Alumni Management

- Verify Offers: Validate rules ("One Student One Job" and "Dream Offer Upgrade") and release offers.
 - Alumni Sync: Auto-onboard graduating students into referral networks.
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6.1 Training & Placement Recruitment Diagram

This diagram illustrates the structured 11-phase operational recruitment process handled by the Training & Placement Cell.



T&P Cell Recruitment Diagram (Image missing)

7. Library Admin & Operations Workflow

The Library Management module automates item inventories, physical space mapping (racks/shelves), ID barcode generations, book loans, returns, and digital library resources.

Phase 1: Library Setup & Policies

- Configuration: Define loan limits (e.g., max 3 books), overdue fine rates, and academic calendars.
- Directory Mapping: Create subjects, authors, and classification hierarchies.
- Rack & Shelf Mapping: Catalog physical locations (e.g., Rack B, Shelf 4) to enable search.

Phase 2: Inventory Cataloging

- Book Intake: Input details (ISBN, title, author, edition).
- Barcoding: Auto-generate barcodes/QR codes to label book spines.

Phase 3: Member Registration

- Verification: Auto-fetch student/faculty status from master DB.
- ID Printing: Print library cards containing barcodes for fast scanner checkouts.

Phase 4: Book Issue & Return Workflow

- Scanner Checkout: Scan student card, check limits/fines, scan book barcode.
- Transaction Log: Record transaction and compute custom due date.
- Scanner Return: Calculate overdue days and assess book condition (Good, Damaged, Lost) and return book to stock.

Phase 5: Fine Management & POS Checkout

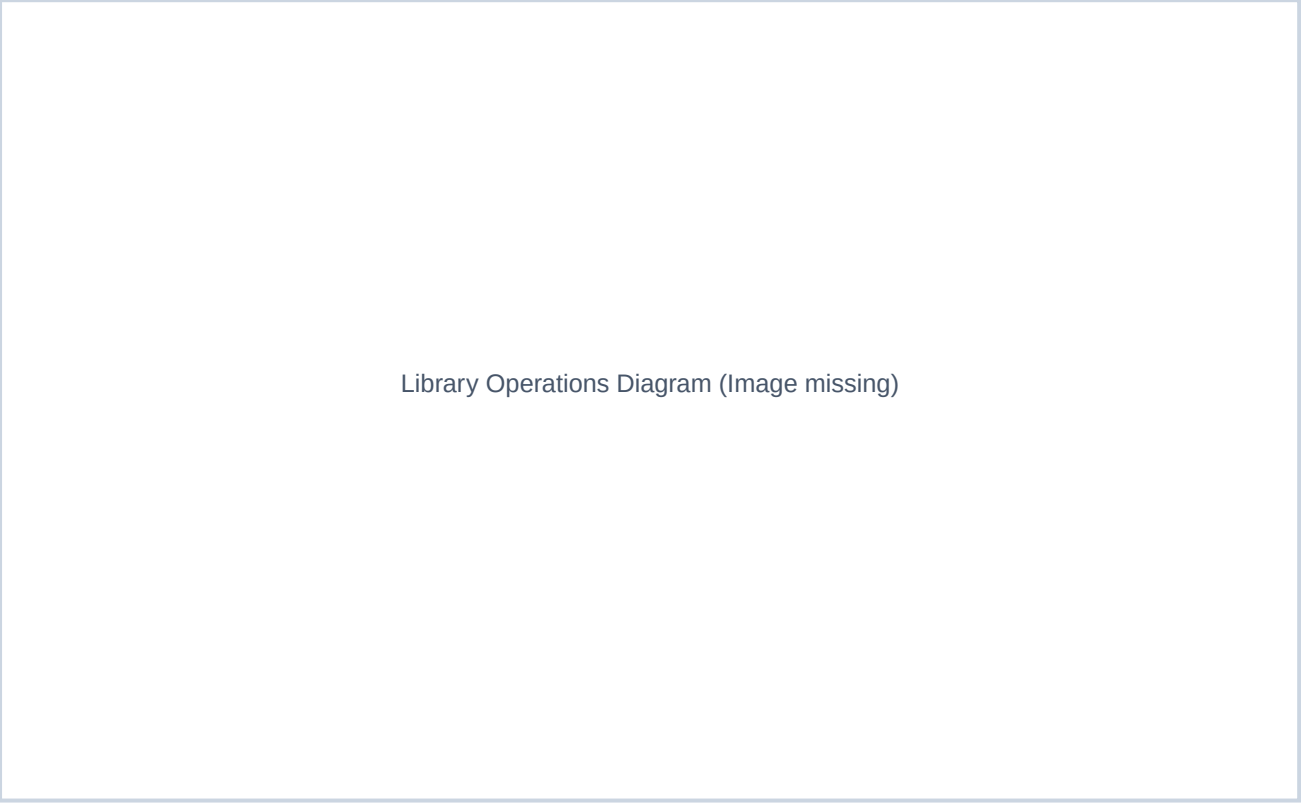
- POS Checkout: Support overdue fee payment via cash or integrated UPI, instant print receipt and ledger sync.

Phase 6: Digital Library Hub & Notifications

- Upload: Store PDFs, lecture notes, academic journals matched to semester rules.
 - Reminders: Auto-sends emails/notifications for books due tomorrow or overdue warnings.
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7.1 Library Admin & Operations Diagram

This diagram displays the 11-phase workflow cataloging physical books, scanner checkouts/returns, POS fine collection, and digital uploads.



Library Operations Diagram (Image missing)

